TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. Which of the following organisms is correctly paired with a trophic level?
   (A) cyanobacterium—secondary consumer
   (B) grasshopper—primary consumer
   (C) zooplankton—primary producer
   (D) grass—decomposer

2. Which of these ecosystems has the *lowest* net primary production per square meter?
   (A) a salt marsh
   (B) an open ocean
   (C) a coral reef
   (D) a tropical rain forest

3. The discipline that applies ecological principles to returning degraded ecosystems to a more natural state is known as
   (A) restoration ecology.
   (B) thermodynamics.
   (C) eutrophication.
   (D) biogeochemistry.

Levels 3-4: Applying/Analyzing

4. Nitrifying bacteria participate in the nitrogen cycle mainly by
   (A) converting nitrogen gas to ammonia.
   (B) releasing ammonium from organic compounds, thus returning it to the soil.
   (C) converting ammonium to nitrate, which plants absorb.
   (D) incorporating nitrogen into amino acids and organic compounds.

5. Which of the following has the greatest effect on the rate of chemical cycling in an ecosystem?
   (A) the rate of decomposition in the ecosystem
   (B) the production efficiency of the ecosystem's consumers
   (C) the trophic efficiency of the ecosystem
   (D) the location of the nutrient reservoirs in the ecosystem

6. Which of the following was a result of the Hubbard Brook watershed deforestation experiment?
   (A) Most minerals were not recycled within the intact forest ecosystem.
   (B) Calcium levels remained high in the soil of deforested areas.
   (C) Deforestation decreased water runoff.
   (D) The nitrate concentration in waters draining the deforested area became dangerously high.

7. Which of the following is an example of bioremediation?
   (A) adding nitrogen-fixing microorganisms to a degraded ecosystem to increase nitrogen availability
   (B) using a bulldozer to regrade a strip mine
   (C) reconfiguring the channel of a river
   (D) adding seeds of a chromium-accumulating plant to soil contaminated by chromium

8. If you applied a fungicide to a cornfield, what would you expect to happen to the rate of decomposition and net ecosystem production (NEP)?
   (A) Both decomposition rate and NEP would decrease.
   (B) Neither would change.
   (C) Decomposition rate would increase and NEP would decrease.
   (D) Decomposition rate would decrease and NEP would increase.

Levels 5-6: Evaluating/Creating

9. **DRAW IT** (a) Draw a simplified global water cycle showing ocean, land, atmosphere, and runoff from the land to the ocean. Label your drawing with these annual water fluxes:
   - ocean evaporation, 425 km^3
   - ocean evaporation that returns to the ocean as precipitation, 385 km^3
   - ocean evaporation that falls as precipitation on land, 40 km^3
   - evapotranspiration from plants and soil that falls as precipitation on land, 70 km^3
   - runoff to the oceans, 40 km^3

   (b) What is the ratio of ocean evaporation that falls as precipitation on land compared with runoff from land to the oceans? (c) How would this ratio change during an ice age, and why?

10. **EVOLUTION CONECTION** Some biologists have suggested that ecosystems are emergent, "living" systems capable of evolving. One manifestation of this idea is environmentalist James Lovelock's Gaia hypothesis, which views Earth itself as a living, homeostatic entity—a kind of superorganism. Are ecosystems capable of evolving? If so, would this be a form of Darwinian evolution? Why or why not? Explain.

11. **SCIENTIFIC INQUIRY** Using two neighboring ponds in a forest as your study site, design a controlled experiment to measure the effect of falling leaves on net primary production in a pond.

12. **WRITE ABOUT A THEME: ENERGY AND MATTER** Decomposition typically occurs quickly in moist tropical forests. However, waterlogging in the soil of some moist tropical forests results over time in a buildup of organic matter called "peat." In a short essay (100–150 words), discuss the relationship of net primary production, net ecosystem production, and decomposition for such an ecosystem. Are NPP and NEP likely to be positive? What do you think would happen to NEP if a landowner drained the water from a tropical peatland, exposing the organic matter to air?

13. **SYNTHESIZE YOUR KNOWLEDGE**

This dung beetle (genus *Scarabaeus*) is burying a ball of dung it has collected from a large mammalian herbivore in Kenya. Explain why this process is important for the cycling of nutrients and for primary production.

For selected answers, see Appendix A.